

Planar Canonical-Lift Diagrams and Reusable Sector Kernels

for Unknot Recognition

Complete matching-dimension-three enumeration, independent coverage certificates,
and a growing geometric family

Prepared for Vladimir Reshetnikov and the ProveIt project
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Abstract

We extend the preceding minimum-envelope method from matching nullity at most two to at most three. For one supplied quadrilateral sector with k allowed types, contraction of zero-labelled face equations leaves at most $8k$ triangle classes. Its canonical standard-coordinate lift is linear on the common refinement of planar minimum diagrams. We prove the explicit output bound

$$R \leq b + 2 \sum_v (m_v - 1) + 9 \sum_{v < w} (m_v - 1)(m_w - 1) = O(k^2),$$

including degeneracies and a linear bound when only one vertex group has a nontrivial diagram. A simple exact implementation constructs all rays in $O(k^3 + tk^2)$ scalar operations after matching elimination, where t is the number of tetrahedra; integer normalization and bit length are charged separately. We also prove completeness of a compact certificate based on dominance inequalities and exact area or length, and reduce reusable pre-elimination sector construction to $O(t \log(t + 1) + k^2)$ elementary integer work.

A double-capped Fibonacci solid-torus family has exactly seven nonlink standard rays for every size, four representing essential discs; the old full arrangement generates quadratically many positive candidates. An abstract two-group family proves that quadratic output is unavoidable for the general potential-cone model. Exact differential checks cover 9,807 sectors on 48 triangulations, including 463 newly eligible sectors of nullity three. The accompanying code, replay certificates, frozen oracle, paired timings, and additive patch make the results reproducible. These are local algorithmic theorems. A complete quasi-polynomial recognizer still requires a proved bound on sector discovery, intermediate states, and all topology transitions.

Repository baseline: <https://github.com/VladimirReshetnikov/ProveIt>

Commit 8188525b70033dcfe7c51ea5ae2c8723ad0c0198. This package proposes additive research modules; it does not change the maintained recognizer's dispatch.

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1 Question, scope, and relation to the existing programme

The immediate problem is to make complete normal-surface search less expensive without weakening its mathematical meaning. A fast test of one supplied surface, a fast search for one favourable linear objective, and a fast enumeration of every relevant vertex surface are different operations. An improvement in the first does not automatically improve the worst-case complexity of the third, and none by itself bounds the number of triangulations or sectors that a recognizer must visit.

The present continuation targets a specific expense exposed by the preceding report: once matching has reduced a sector to a small quadrilateral subspace, the existing general arrangement routine still intersects equality hyperplanes that never change the canonical lift. In dimension three, the normalized feasible set is a polygon. The relevant geometry is therefore the overlay of the actual planar regions on which each triangle potential attains its minimum. Enumerating those regions gives both a stronger output bound and a direct certificate that the whole supplied sector has been covered.

The second expense is earlier in the pipeline. The original constructor materializes a length- k quadrilateral label for every face-corner equation, although all but $O(k)$ labels vanish. Compiling the source incidences once and postponing dense labels until after this test removes that tk allocation pattern. These two changes address separate stages, and the experiments time them separately.

1.1 The object actually being enumerated

Let T be a finite compact triangulation with t tetrahedra. Normal coordinates use four triangle counts and three quadrilateral counts in each tetrahedron. A *supplied sector* chooses at most one allowed quadrilateral type per tetrahedron and sets all other quadrilateral coordinates to zero. Its number of allowed types is $k \leq t$. Allowed coordinates may vanish: a sector contains all its coordinate faces. The remaining matching equations and nonnegativity inequalities define a rational polyhedral cone in standard coordinates. We enumerate its extreme rays with nonzero quadrilateral part, in primitive integer normalization. The excluded rays are pure vertex links.

This is a complete list of the nonlink *standard* rays of the sector. It is not merely a list of extreme rays in quadrilateral coordinates. Canonical lifting is positively homogeneous but piecewise linear, so a quadrilateral direction in the interior of a quadrilateral face can lift to a standard extreme ray. This distinction is established in the classical conversion theory [1] and is essential here: the geometric family in Section 6 has four essential standard disc rays but only one of them is quadrilateral-extreme.

The abstract cone arguments apply whenever the stated potential model is available. The executable modules use the maintained native validator, whose contract is narrower: a connected orientable finite 3-manifold triangulation with one torus boundary component. They do not implement ideal-vertex or arbitrary boundary-pattern input. The matching dimension d is the dimension of the rational quadrilateral matching kernel before nonnegativity. The implemented eligibility test is $d \leq 3$. Nonnegativity may reduce the feasible dimension further; the empty, point, segment, and polygon cases all occur in genuine audited sources.

1.2 What is new in this continuation

The main mathematical statements are as follows.

1. **Reusable sparse construction.** Theorem 2.2 proves a pre-elimination work and storage bound while preserving the incumbent kernel’s exact combinatorial representatives, potential gauge, cycle rows, and rational basis.
2. **Complete planar enumeration.** Theorem 3.5 gives a bound in terms of the *active* minimum cells, rather than all pairwise potential equalities. Proposition 3.7 gives an exact polynomial construction and accounts for writing the full normal vectors.
3. **Independent completeness replay.** Theorems 4.3 and 4.5 show why dominance and exact total area suffice to certify maximal minimum cells, not just a partial collection of valid rays.
4. **An infinite geometric test family.** Section 6 solves the entire cone and topology of the two-cap Fibonacci family, including the exact growth of the old arrangement’s rejected directions. Section 5 separately shows quadratic sharpness for abstract potential cones.

The canonical-lift characterization itself is not a new claim. It is proved here to fix every degeneracy needed by the algorithm, with attribution to both the earlier report and the classical conversion framework. Planar minimum diagrams are clipped power diagrams; their computational geometry and the quadratic complexity of three-dimensional normal-fan overlays have substantial prior literature [2, 3, 4]. Our contribution is the explicit complete normal-sector algorithm, the parameter-sensitive boundary count, its independent certificate, the exact implementation, and the geometric family. We make no claim to a first general theorem about planar subdivisions.

1.3 Repository lineage

All code and incoming reports were read against the pinned commit

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The immediately preceding minimum-envelope package proves complete enumeration for $d \leq 2$ and the general common-cell characterization [10]. Its optimized rank-two implementation is retained unchanged as a performance control. The other recent incoming packages cover different parts of the programme: certified active-face reduction and batched local descent; bounded trace search; boundary assembly kernels; rooted disc components; and affine orbit profiles [11, 12, 13, 14, 15]. Their bounds are not imported as global coverage or termination theorems for this algorithm. Likewise, global admissible-face bounds proved for closed triangulations [5] are not transferred here to finite exteriors with boundary.

In particular, the phrase *cycle kernel* is potentially misleading across the reports. Here it means linear consistency equations around cycles in the graph of triangle-coordinate differences. The cycle-envelope incoming report studies a boundary assembly language. These are distinct objects with distinct parameters.

The current maintained source already includes a native, certified diagram-to-exterior route; its pulling subdivision constructs $20 \max(1, n)$ tetrahedra from an n -crossing diagram. We do not describe source provenance as an absent component. The remaining question is how to use such a source, complete local searches, and certified topology transformations with a sufficiently small *total* search bound. The present modules accept a supplied triangulation and do not independently authenticate an accompanying knot diagram.

1.4 The asymptotic target

We use *quasi-polynomial* for time $2^{(\log n)^{O(1)}}$. The more specific targets $n^{O(\log n)}$ and $n^{O((\log n)^2)}$ are different. Lackenby’s March 2021 handout announces

$$2^{O((\log n)^3)} = n^{O((\log n)^2)}$$

for unknot recognition [8]. It describes a hierarchy strategy combining controlled surface complexity, compressed representations, multisurfaces, and shortened hierarchy depth. The July 2026 manuscript gives detailed algorithms and certification theorems for incompressible surfaces and essential hierarchies, while explicitly leaving speed-up for further work [9]. We therefore use those sources as structural guidance, without attributing a fully implemented quasi-polynomial analysis to that manuscript or to the present code.

For comparison, Burton and Ozlen’s branching algorithm supplies a rigorous exponential worst-case analysis and strong experimental behaviour [6]. Improving one subroutine here neither changes that theorem nor establishes a polynomial empirical law for complete recognition. The statements below are local theorems and reproducible stage measurements; Section 9 specifies what would be needed to compose them into a global bound.

2 Reusable sparse construction of the matching sector

This section proves the construction bound used by the implementation. It changes representation and reuse, not the normal matching equations. For a paired face and one of its three corners, matching has the form

$$X_a + Q_\alpha = X_b + Q_\beta, \quad X_b - X_a = Q_\alpha - Q_\beta.$$

Here X_a, X_b are triangle coordinates and the two quadrilateral incidences have coefficients $+1, -1$. If the two global types coincide, their contributions cancel. The source can store this equation as its two corner indices and at most two signed type identifiers, before any sector is chosen.

Lemma 2.1 (Incidence and retained-class bounds). *For a supplied sector with k allowed quadrilateral types, at most $4k$ face-corner equations have a nonzero quadrilateral label. After all zero-labelled equations are contracted, the total number p of triangle classes in nonsingleton vertex groups is at most $8k$. Every coefficient of a path potential, and every coefficient of an unnormalized fundamental-cycle row, has absolute value at most four.*

Proof. A quadrilateral has one normal arc in each of its four faces. It therefore occurs in at most four paired face-corner equations, counting incidences before any cancellation. The total number of active signed incidences is at most $4k$; each nonzero labelled equation contains at least one.

Triangle corners around a fixed vertex are connected by the original face-corner graph. Contracting its zero-labelled edges preserves connectedness. Every class in a nonsingleton quotient component is incident with an active edge. The number of these classes is at most twice the number of active edges, hence at most $8k$. Singleton components may be numerous, but contribute no retained triangle-difference variable.

Choose a spanning forest of the quotient graph. A path from a root to a class traverses each forest edge at most once. A fundamental cycle consists of one nonforest edge and the simple forest path between its endpoints. Each original signed incidence can therefore contribute at most once to either sum. Each allowed quadrilateral has at most four such incidences in the entire graph, proving the coefficient bound. Dividing a nonzero cycle row by the gcd of its entries cannot increase it. $\square$

The last bound is about coefficients of linear forms, not the values of normal coordinates. The kernel basis can have large rational coefficients, and a primitive normal vector can have exponentially large integer coordinates. No unit-cost assumption about those values is used.

Theorem 2.2 (Sparse pre-elimination constructor). *A validated source can be compiled into $O(t)$ constant-size face-corner incidence records. For each allowed support, a constructor can produce the incumbent retained matrix, canonical corner classes, vertex groups, path potentials, and lexicographically ordered primitive cycle rows in*

$$O(t \log(t+1) + k^2)$$

elementary integer operations, using $O(t + k^2)$ integer slots. The indices in those slots require $O(\log(t+1))$ bits. The cost of exact elimination and its resulting basis is additional. The same bounds include one-time native validation and compilation, using balanced union structures; subsequent sectors can reuse them.

Proof. Scan the compiled incidences and keep only selected types. Zero labels are recognized from a record of length at most two, without allocating a length- k vector. Contract their corner pairs using union by size. Each parent path has length $O(\log(t+1))$, since the containing component at least doubles whenever the root is attached below another root. Path compression can only shorten these paths. A separate minimum-corner field gives every component its least original corner index, independent of the balancing choices. This is the canonical representative used by the original constructor, whose structural union root need not be retained.

There are $O(t)$ equations and corners. Computing representatives, grouping by the unchanged source vertex identities, and sorting canonical representatives costs $O(t \log(t+1))$. The support dictionary has k selected entries. It may be implemented by balanced search or sorted lookups within this bound; its index values specify the incumbent column order.

Only the at most $4k$ active equations now receive dense labels. Each label and each retained matching row has $O(k)$ entries, because $p \leq 8k$. The total materialization cost is $O(k^2)$. The quotient adjacency lists preserve the original scan order. Breadth-first traversal in the original root order consequently chooses the same forest and the same potential gauge as the old routine. Every traversed active edge requires $O(k)$ additions. Isolated components all reference one immutable zero tuple, so their number contributes $O(t)$ pointers rather than $O(tk)$ zero entries.

There are $O(k)$ fundamental cycle rows of length k . Producing and normalizing them costs $O(k^2)$, with bounded coefficients by Lemma 2.1. To preserve the old lexicographic order without an extra logarithmic factor from long tuple comparisons, apply stable radix sorting from the last column to the first over the fixed alphabet $\{-4, \dots, 4\}$. Sorting and deleting adjacent duplicates costs $O(k^2)$. The supplied implementation first deduplicates with a set and then performs this radix sort; the deterministic comparison-free version just described proves the bound without relying on hashing.

Native validation uses a linear inventory of face, corner, and edge incidences together with balanced signed union structures and linear topological checks. This fits $O(t \log(t+1))$ elementary work. Compilation then adds only a linear scan. The listed arrays contain $O(t)$ source and quotient records and $O(k^2)$ dense entries, proving the storage bound. Rational elimination has not yet been performed and is charged separately. $\square$

2.1 Compatibility and the implementation contract

The constructor returns the existing `SectorKernel` type. The source's validation data, support ordering, minimum corner identities, adjacency order, canonical potentials, primitive cycle rows,

and exact reduced nullspace agree with the incumbent routine. The row space being equal would suffice mathematically, but exact field agreement is a stronger regression check and prevents accidental changes to deterministic output or certificate labelling. The implementation deliberately calls the same dense rational elimination routine after the smaller allocation phase. We therefore claim no improvement to the asymptotic cost of that elimination.

The direct implementation uses Python dictionaries and sets for indexing and deduplication. The displayed deterministic bound is for the balanced-index and radix-sort algorithm in the proof. The usual expected constant-time hashing model gives the corresponding bound for those Python operations. Adversarial hash collisions can worsen the Python container costs; they do not invalidate the algebra or the polynomial bit-complexity conclusion. The numerical experiments measure the actual Python implementation, not the idealized unit-cost model.

`PreparedSectorSource` validates one supplied triangulation and captures its source digest. It can build many kernels from that same source. The source dictionary, the prepared object, and the `prepared` field shared by returned kernels have an explicit read-only contract. They are not deep-frozen copies. The certificate-producing API compares the captured digest with the supplied source, but callers must still avoid mutating shared validation data. Abandoning an interrupted sector build leaves the prepared source reusable.

This distinction explains the timing design. A one-shot sparse constructor still pays native validation and compilation; it can be slower than the original routine on a small instance. Repeated sector work can amortize those costs. Comparisons against an original constructor supplied with already prepared geometry are needed to distinguish the allocation improvement from validation reuse.

3 Minimum diagrams and complete planar sector enumeration

3.1 The retained cone and its canonical lift

Fix one admissible quadrilateral sector. Let t be the number of tetrahedra and k the number of allowed quadrilateral coordinates. The support reduction supplies a rational matching subspace $K \subseteq \mathbb{R}^k$, together with retained triangle classes partitioned into vertex groups I_v . Write $p_v = |I_v|$ and $p = \sum_v p_v$. The bound $p \leq 8k$ is a property of that reduction; the geometric arguments below first keep p as a separate parameter. Discarded independent vertex-link factors can be reattached afterwards. They add their own link rays and do not change the nonlink rays considered here.

For each $c \in I_v$, a rational linear potential $\pi_{v,c} : K \rightarrow \mathbb{R}$ records the triangle-coordinate differences imposed by matching. In the actual construction the potentials have integer coefficients, obtained by summing signed quadrilateral labels along paths. Most of the geometry only needs rational coefficients. Put

$$\mathcal{Q} = K \cap \mathbb{R}_{\geq 0}^k, \quad \mathcal{C} = \left\{ (q, x) : \begin{array}{ll} q \in \mathcal{Q}, & x_{v,c} = \pi_{v,c}(q) + a_v \quad (c \in I_v), \\ a_v \in \mathbb{R}, & x_{v,c} \geq 0 \end{array} \right\}.$$

The scalar a_v is shared within its group. Let ℓ_v be the vector with zero quadrilateral part, entries one in group v , and zero in the other groups. Define

$$\mu_v(q) = \min_{c \in I_v} \pi_{v,c}(q), \quad \mathcal{F}(q) = \left(q, (\pi_{v,c}(q) - \mu_v(q))_{v,c} \right). \quad (1)$$

This is the canonical triangle normalization used in quadrilateral-to-standard conversion: each vertex group has at least one zero triangle coordinate. Its role in normal-coordinate conversion is established in Burton's work [1].

Lemma 3.1 (Canonical decomposition). *Every $z = (q, x) \in \mathcal{C}$ has a unique expression*

$$z = \mathcal{F}(q) + \sum_v b_v \ell_v, \quad b_v \geq 0.$$

The map $\mathcal{F}$ is positively homogeneous and injective.

Proof. If $x_{v,c} = \pi_{v,c}(q) + a_v$, nonnegativity is equivalent to $a_v + \mu_v(q) \geq 0$. Set $b_v = a_v + \mu_v(q)$. Subtraction gives the asserted expression. The coefficients are unique because they are the common differences within disjoint groups; equivalently $b_v = \min_{c \in I_v} x_{v,c}$. The minimum of finitely many linear forms is positively homogeneous for nonnegative scalars, proving homogeneity of $\mathcal{F}$. Its first component is q , proving injectivity. $\square$

Set $s(q) = \sum_{i=1}^k q_i$ and

$$P = \{q \in \mathcal{Q} : s(q) = 1\}. \quad (2)$$

If $\mathcal{Q} \neq \{0\}$, this is a nonempty compact rational polytope: it is a closed subset of the standard simplex, and every nonzero nonnegative q has $s(q) > 0$. If $d = \dim K \leq 3$, then the effective dimension $r = \dim P$ is at most two. It may be smaller than $d - 1$. An empty section, a point, and a segment must consequently be handled as genuine possibilities even when the matching nullity is three.

Restrict each potential to $\text{aff}(P)$. Identical restrictions are identified for geometric purposes, without deleting any output triangle coordinate. For a group define its closed minimum cells by

$$C_{v,c} = P \cap \bigcap_{a \in I_v} \{\pi_{v,c} \leq \pi_{v,a}\}. \quad (3)$$

Take the full-dimensional cells, relative to $\text{aff}(P)$, and all their faces. Denote this minimum subdivision by $\mathcal{D}_v$, and denote their common refinement by $\mathcal{D}$. In particular, the faces of P are present in these complexes. Let m_v be the number of full-dimensional cells of $\mathcal{D}_v$.

Lemma 3.2 (Degenerate minimum labels). *The full-dimensional minimum cells cover P , meet face-to-face, and satisfy $1 \leq m_v \leq p_v$. A potential whose minimum cell has smaller dimension introduces no additional geometric subdivision of these cells and their faces.*

Proof. Outside the finite collection of equality hyperplanes of distinct restricted forms, the minimizing restricted form is unique. Such points are dense in the relative interior of P . Approximating any point of P by them and taking a subsequence with a fixed minimizing label proves coverage by closures of full-dimensional cells. Distinct such cells have disjoint relative interiors. Moreover

$$C_{v,c} \cap C_{v,a} = C_{v,c} \cap \{\pi_{v,a} - \pi_{v,c} = 0\}.$$

The difference on the right is nonnegative on $C_{v,c}$; thus the intersection is an exposed face of that cell, and symmetrically of the other cell.

Suppose now that c is a label with a smaller-dimensional minimum cell. On any full-dimensional cell with minimizer a , the affine gap $\pi_{v,c} - \pi_{v,a}$ is nonnegative. Its zero set there is a face. More generally, on the relative interior of any face, an affine nonnegative function can vanish only everywhere on that face or nowhere in its relative interior. Indeed, if it vanishes at a relative interior point, its linear part must vanish in all directions parallel to the face, since short displacements in both directions remain in the face. Hence the extra label cannot split an existing face into smaller pieces. The one-dimensional and zero-dimensional versions of this argument are included. $\square$

The common-cell extreme-ray characterization below is the general minimum-envelope theorem of the preceding repository report [10]. A proof is included to make explicit the treatment of active quadrilateral coordinates and degenerate minimum labels on which the present complete planar algorithm relies.

Theorem 3.3 (Common-cell characterization). *The extreme rays of $\mathcal{C}$ are its link rays $\mathbb{R}_{\geq 0}\ell_v$ and the rays $\mathbb{R}_{\geq 0}\mathcal{F}(q)$ indexed by vertices q of $\mathcal{D}$. Different vertices give different nonlink rays.*

Proof. The face with $q = 0$ is the nonnegative span of the link vectors. Since their supports are disjoint, its extreme rays are precisely the link rays. It is a face of the whole cone: if nonnegative quadrilateral parts sum to zero, both are zero. For $q \neq 0$, any positive link coefficient in Lemma 3.1 gives a decomposition into a nonzero canonical vector and a nonzero vector with zero quadrilateral part. These cannot be proportional. A nonlink extreme ray must therefore have its canonical representative.

Normalize $s(q) = 1$. On each cell of $\mathcal{D}$, a fixed minimizing form may be chosen in every group, so $\mathcal{F}$ is affine on that cell. If q is not a vertex, it lies in the relative interior of a positive-dimensional face. There are distinct nearby points q^+, q^- of that face with $q = (q^+ + q^-)/2$. Therefore

$$\mathcal{F}(q) = \frac{1}{2}\mathcal{F}(q^+) + \frac{1}{2}\mathcal{F}(q^-).$$

The two vectors cannot be proportional, because their quadrilateral parts are distinct and both have sum one. Thus the ray is not extreme.

Conversely, let q be a vertex of $\mathcal{D}$. Choose a maximal common cell containing it and one defining minimizer c_v for each group. Suppose $\mathcal{F}(q) = z_1 + z_2$ with $z_i \in \mathcal{C}$. The coordinate x_{v,c_v} is zero in $\mathcal{F}(q)$. Nonnegativity forces the same coordinate to be zero in each summand. Thus c_v minimizes in both summands, both summands are canonical, and their quadrilateral parts q_1, q_2 belong to the same chosen minimum cones. A summand with $q_i = 0$ is zero: its zero coordinate in each group forces every link coefficient to vanish.

If both summands are nonzero, set $\lambda_i = s(q_i) > 0$. Then $\lambda_1 + \lambda_2 = 1$, and

$$q = \lambda_1(q_1/\lambda_1) + \lambda_2(q_2/\lambda_2).$$

Both normalized points belong to the selected common cell. A vertex of a polyhedral complex is a vertex of every cell containing it, so both normalized points equal q . Positive homogeneity makes $z_i = \lambda_i\mathcal{F}(q)$, as required. Working inside P has included all active inequalities $q_i = 0$; they cannot be omitted from this extremality argument. Finally, the normalization $s(q) = 1$ and the quadrilateral projection distinguish the represented rays. $\square$

3.2 A planar count with the common boundary counted once

Assume for this subsection that P is two-dimensional, and let b be its number of genuine polygon corners. Then $b \leq k$. We use polygon representations with no repeated vertex and no collinear consecutive vertices. In a subdivision graph, removable collinear degree-two points are suppressed. The following count is useful because its bound for *internal* edges is independent of b .

Lemma 3.4 (One-group planar count). *For a minimum subdivision with m full-dimensional cells, let V_i, V_b be its numbers of interior and boundary vertices, and E_i its number of internal edges. Then*

$$V_i + V_b - b \leq 2(m - 1), \quad E_i \leq 3(m - 1). \quad (4)$$

The total number of polygon-vertex incidences over all its cells is at most $b + 8(m - 1)$.

Proof. Include the boundary cycle of P in the finite embedded graph, and let $c \geq 1$ be its number of components. There are $m + 1$ faces, including the exterior face, and the number of boundary edges is V_b . Euler's formula gives

$$V_i + V_b - (E_i + V_b) + (m + 1) = 1 + c, \quad E_i = V_i + m - c.$$

Every interior vertex has degree at least three. There cannot be a degree-one interior endpoint in a face-to-face cover by full-dimensional polygons. If an interior vertex had degree two, the two incident full-dimensional cells would share their affine equality line through it; their two incident edges would be collinear and the vertex removable. Likewise a boundary vertex other than a genuine corner has degree at least three, or it would be removable. The b corners have degree at least two. Summing degrees yields

$$2(E_i + V_b) \geq 3V_i + 3(V_b - b) + 2b, \quad 2E_i \geq 3V_i + V_b - b.$$

Substituting the Euler identity proves

$$V_i + V_b \leq 2m - 2c + b.$$

Since $c \geq 1$, this proves the first asserted bound. Since $V_b \geq b$, it also gives $V_i \leq 2m - 2c$; the Euler identity now gives $E_i \leq 3m - 3c$.

An internal edge is counted twice in the boundaries of the cells, and a boundary edge once. Thus the total incidence count is $2E_i + V_b$, which is at most $6(m - 1) + b + 2(m - 1)$. This proof does not require a separate connectivity argument for the subdivision graph. $\square$

Put $a_v = m_v - 1$ and $A = \sum_v a_v$. Let R denote the number of nonlink rays, equivalently the number of vertices in the common subdivision.

Theorem 3.5 (Planar output bound). *If $\dim P = 2$, then*

$$R \leq b + 2A + 9 \sum_{v < w} a_v a_w. \quad (5)$$

The numbers of vertices, edges, and faces of the common subdivision are all $O(k + p^2)$. For one nontrivial vertex group the sharper bound $R \leq b + 2(m_v - 1)$ holds.

Proof. The union of the polygon corners and the individual group vertices contains at most $b + 2A$ points by Lemma 3.4. Every other overlay vertex is an intersection of internal edges from different groups. Two noncollinear segments meet in at most one point. Collinear overlapping segments produce no new vertex except an endpoint of one of the original segments, already in the preceding union. Group v has at most $3a_v$ internal edges, so the additional vertex count is at most $\sum_{v < w} (3a_v)(3a_w)$. Theorem 3.3 transfers this count to rays.

There are at most $3A$ internal segments before overlay. Each can be split at only $O(A)$ intersections and endpoints of other segments, including collinear overlaps. Consequently the overlay has $O(A^2)$ internal edges. Its boundary has $b + O(A)$ edges, because each original internal segment contributes at most two boundary endpoints. Euler's formula bounds the number of faces by the same asymptotic quantity. If there is only one nontrivial group, the intersection term in (5) vanishes. $\square$

The distinction between internal edges and the common boundary is essential in this proof. Summing an $O(k + p_v)$ bound for whole group diagrams repeats the same boundary for each group and does not by itself provide an $O(k + p)$ collection of overlay segments. Lemma 3.4 supplies the needed $O(p_v)$ internal bound.

Corollary 3.6 (Effective dimensions zero and one). *If P is a segment, then*

$$R \leq 2 + \sum_v (m_v - 1).$$

If P is a point, $R = 1$; if P is empty, $R = 0$. Thus every supplied sector of matching nullity at most three has $O(k + p^2)$ nonlink rays, and the retained normal-sector bound $p \leq 8k$ makes this $O(k^2)$.

Proof. On a segment, every minimum cell is an interval. Distinct full-dimensional intervals have disjoint interiors, and a group with m_v such intervals contributes at most $m_v - 1$ interior breakpoints. The common subdivision uses their union and the two section endpoints. The other cases follow directly from Theorem 3.3. $\square$

These minimum diagrams belong to established computational geometry. For $f_i(y) = u_i \cdot y + \beta_i$, put $z_i = -u_i/2$ and $w_i = \|u_i\|^2/4 - \beta_i$. Then

$$\|y - z_i\|^2 - w_i = \|y\|^2 + f_i(y).$$

The common quadratic term does not affect minimizers, so each group diagram is a clipped power diagram [2]. The present count is a direct planar argument adapted to the normal-sector lift, rather than a new general theorem about power diagrams.

3.3 Exact construction and the cost of writing the rays

One complete construction first obtains P in rational affine coordinates. If the section is two-dimensional, for each distinct restricted group form f_c it computes

$$P \cap \bigcap_a \{f_a - f_c \geq 0\}$$

by successive exact half-plane clipping. Empty or lower-dimensional results can be discarded immediately: further intersections cannot restore their dimension. Positive-area cells are recorded with their genuine corners. Their edges on ∂P are omitted from the internal segment set; the remaining edges are deduplicated. The candidate vertices are the union of section corners, cell corners, and intersections of internal edges from different groups. All predicates are exact and all clipping inequalities are closed. Multiple ties, parallel edges, coincident intersections, and overlaps therefore require no generic perturbation.

For a segment, restrict the forms to a parameter $u \in [0, 1]$. The lower envelope of lines can be constructed by sorting slopes and maintaining increasing breakpoint positions. Equal slopes retain the lowest intercept, and equal affine forms retain one representative. Zero-length pieces are omitted, while the endpoints of positive-length pieces are retained. A point requires only one canonical lift. These branches use the feasible dimension, not merely the dimension of the matching kernel.

Proposition 3.7 (Constructive operation bound). *After the sector kernel has been constructed, complete geometry and all native nonlink ray vectors can be obtained using*

$$O(k^3 + kp^2 + p^3 + (k + p)(k + p^2) + t(k + p^2))$$

exact scalar arithmetic and comparison operations, with lower-order terms absorbed. Under $p \leq 8k$, this is

$$O(k^3 + tk^2). \tag{6}$$

Integer gcd and lcm computations for primitive normalization are additional operations on integers of polynomial bit length.

Proof. In an affine coordinate space of dimension at most two, one may construct a bounded section by testing pairwise intersections of the k boundary lines against all k inequalities and taking their exact convex hull. This elementary fallback costs $O(k^3)$ operations and handles empty, segment, and point sections as well. The implementation can instead clip a justified coordinate box when its chart uses actual q -entries.

A polygon initially having $b \leq k$ sides has at most $b + j$ sides after j half-plane clips: a clip replaces a consecutive boundary chain by at most one new side. Thus computing every cell of group v takes $O(p_v^2(k + p_v))$ operations. Summing gives $O(kp^2 + p^3)$. There are only $O(p)$ distinct internal segments, so their pairwise intersection tests cost $O(p^2)$. Exact comparison sorting for deduplication stays within the displayed bound.

Project all retained potentials into the affine chart once. At each of $O(k + p^2)$ resulting points, evaluating the quadrilateral forms and triangle potentials and taking the group minima then costs $O(k + p)$, rather than a fresh length- k dot product for every potential. Expanding repeated triangle classes to the native $7t$ -coordinate vector, and checking its native matching equations, costs $O(t + k)$ per ray. These charges prove the assertion. $\square$

The quadratic ray count is not a quadratic bound for explicit native output. The geometry uses $O(k + p^2)$ candidate points, but writing R native vectors requires $\Theta(tR)$ coordinate positions in this representation. In the normal-sector regime, the output alone can therefore have $O(tk^2)$ entries. This distinction is retained in (6).

In the integer-potential model, one may first clear the denominators of the quadrilateral direction and divide its entries by their gcd. Every triangle potential, and then every canonical triangle coordinate, is integral on this primitive integer q . The whole standard vector is primitive because its quadrilateral coordinates already have gcd one. This procedure uses $O(k)$ gcd/lcm steps per ray, besides scalar evaluation and output. For arbitrary rational potentials the corresponding safe procedure clears denominators in the entire lift instead; the ray theorem does not assume that primitive quadrilaterals imply integral triangles.

Let B bound the bit length of the rational affine coefficients after exact coordinate reduction. Every geometric vertex is the intersection of two source boundary or active minimum-equality lines. The constant-size determinant formulas therefore have polynomial, indeed $O(B)$, rational encoding length up to fixed constant factors. Section endpoints obey the analogous one-dimensional bound. The corresponding bound for exact linear algebra does not assume that numerical coefficients remain small as t grows. After clearing row denominators, suppose the integer matrix entries have at most β bits in absolute value. Hadamard's inequality bounds every rank- ρ minor by

$$|\det M| \leq (\sqrt{\rho} 2^\beta)^\rho, \quad \log_2 |\det M| \leq \rho\beta + \frac{\rho}{2} \log_2 \rho$$

when the determinant is nonzero. Exact kernel bases are obtained from ratios of such minors. The normalized chart then uses constant-size inversions because $d \leq 3$; projecting a source potential adds exact products and sums over its k coefficients. These estimates charge both matrix dimensions and coefficient bit lengths, including row denominators. In the generic rational model, these are part of the supplied data. For native matching input, path potentials sum $O(t)$ incidence labels, so their coefficient bit lengths are polynomially bounded before applying the determinant estimate. These determinant bounds consequently give polynomial bit length uniformly for growing t and k , including families with exponentially large primitive coordinates. Retaining support-line descriptions also makes the geometric bounds transparent during clipping. Clearing denominators of k coordinates can produce a common denominator with $O(kB)$ bits in the generic rational model. This remains polynomial, but its arithmetic is not a unit-cost bit operation. Thus the construction has polynomial bit complexity for this fixed dimension, with rational arithmetic and integer normalization costs charged at their actual operand lengths.

Remark 3.8 (Inactive equalities are not minimum edges). It is not correct to treat every equality of two potentials as an edge of the canonical lift. On $P = \{(x, y) : x, y \geq 0, x + y \leq 1\}$, consider $0, 1 + x, 1 + y$. The latter two forms agree on $x = y$, but neither is minimal anywhere. The canonical lift is affine on the whole triangle. The intersection $(1/2, 1/2)$ of that irrelevant equality with the boundary is not an extreme lifted ray; it is a convex combination of the adjacent corner lifts. An arrangement of all pairwise equalities may be used only with a separate extremality filter, and it loses the active-skeleton output bound proved above.

4 Independent certificates of complete minimum coverage

The producer's polygon-clipping sequence need not belong to the verification trust boundary. A compact alternative submits the resulting minimum cells and proves their coverage by exact area or interval length. The checker reconstructs the matching subspace and all native corner potentials independently, so the following statements concern source functions, not functions accepted from the producer.

4.1 Certifying the affine chart and the section

Proposition 4.1 (Chart completeness). *Suppose the source checker establishes $\dim K = d$, where $1 \leq d \leq 3$. A submitted affine chart*

$$q(y) = c_0 + \sum_{j=1}^{d-1} y_j c_j$$

parametrizes all of $K \cap \{s = 1\}$ if the checker verifies $c_j \in K$, $s(c_0) = 1$, $s(c_j) = 0$ for $j > 0$, and distinct coordinate indices $i_1, \dots, i_{d-1}$ satisfying

$$q_{i_j}(y) = y_j.$$

Every feasible chart point then lies in $[0, 1]^{d-1}$.

Proof. The linear map

$$q \mapsto (s(q), q_{i_1}, \dots, q_{i_{d-1}})$$

sends $c_0, c_1, \dots, c_{d-1}$ to the standard basis of $\mathbb{R}^d$. These vectors are linearly independent. Since they belong to the d -dimensional source kernel, they form its basis. A vector in that kernel with $s(q) = 1$ therefore has the asserted affine expression, with coefficients q_{i_j} . For a nonnegative vector of coordinate sum one, each of those actual coordinates belongs to $[0, 1]$. $\square$

If $d = 0$, or if s vanishes identically on K , the nonnegative cone contains only zero and no chart is needed. Otherwise an affine chart is required even when its feasible section later turns out to be empty. Nullity alone cannot justify accepting a missing section.

Proposition 4.2 (Source-facet section certificate). *Let a two-dimensional chart describe the source section by affine inequalities $h_i(y) \geq 0$. A submitted nondegenerate convex polygon S , given by its genuine corners in cyclic order, equals the source section P if the following checks pass: every vertex satisfies all source inequalities, and every edge of S lies on the zero line of a nonconstant source form h_i .*

Proof. Vertex feasibility and affinity give $S \subseteq P$. For an edge of S , its checked source form vanishes on the edge line and is nonnegative on S . The form is nonconstant and S has interior, so it is positive at some point of S . Consequently its nonnegative half-plane is exactly the inward edge half-plane of S . Every point of P lies in that half-plane. Intersecting the inward half-planes for all edges gives $P \subseteq S$, proving equality. $\square$

This two-sided argument does not reconstruct a clipping history. If the submitted section has fewer than three vertices in a two-dimensional chart, the checker instead enumerates all feasible pairwise source-line intersections within the established coordinate box and takes their hull. A nonempty bounded polyhedron in the plane, including a segment or a point, has extreme points determined by independent active constraints, so the fallback is complete. In a one-dimensional chart, direct exact lower and upper endpoint bounds suffice.

4.2 The area and length certificate

Theorem 4.3 (Exact minimum-cell coverage certificate). *Let P be a nondegenerate convex polygon and let $f_1, \dots, f_n$ be its source affine functions after identifying identical restrictions. Suppose a certificate submits finitely many positive-area convex polygons S_i , each carrying one source label, subject to the following conditions:*

1. *no restricted affine function labels two submitted polygons;*
2. *every submitted vertex belongs to P and satisfies $f_i \leq f_j$ for every source function f_j ;*
3. *the sum of submitted polygon areas equals $\text{area}(P)$.*

Then the polygons are exactly the full-dimensional source minimum cells. In particular they cover all of P , including its boundary, and no full-dimensional minimum cell is omitted.

Proof. The vertex checks extend to each polygon by affinity and convexity, so $S_i \subseteq C_i$, where C_i is the true minimum cell. If the relative interiors of polygons carrying different labels intersected, they would intersect in a nonempty open set. On that set both $f_i \leq f_j$ and $f_j \leq f_i$ would hold. The affine difference would vanish on an open set and hence identically, contrary to the distinct-label condition. Thus pairwise intersections have area zero, and

$$\text{area}\left(\bigcup_i S_i\right) = \sum_i \text{area}(S_i) = \text{area}(P).$$

The union is finite and closed. If a point $z \in P$ were missing, some neighborhood of z would be disjoint from the union. Its intersection with P has positive area, even when z is on the boundary. To see this last statement, choose an interior point $w \in P$. Points $(1 - \varepsilon)z + \varepsilon w$, for sufficiently small $\varepsilon > 0$, lie in that neighborhood and in the interior of P ; a sufficiently small disk around one such point lies in both. A missing point therefore contradicts area equality. The polygons cover all of P , not merely an almost-everywhere subset.

Now let C_i be a true full-dimensional minimum cell. Its uniquely minimized points are dense in it: within its interior, remove the finite union of equality lines with the other distinct source forms, and then take the closure. At such a unique-minimizer point, coverage is possible only by a polygon carrying label i . This proves that label i is submitted, and closedness gives $C_i \subseteq S_i$. Together with the already established reverse inclusion it yields $C_i = S_i$. Conversely every submitted polygon has positive area and is contained in its source minimum cell, so that source cell is full-dimensional. These two conclusions prove exact equality of the cell inventories. $\square$

Corollary 4.4 (Interval coverage certificate). *Parametrize a nondegenerate source segment by $u \in [0, 1]$. The corresponding statement holds for positive-length submitted intervals, with pairwise distinct restricted affine labels, endpoint dominance, and total parameter length one.*

Proof. Endpoint dominance holds throughout an interval. Distinct affine functions on a line cannot agree on a nonempty open interval, so submitted interiors are disjoint. Exact total length proves that their finite closed union is $[0, 1]$; any missing point would leave a positive-length relative neighborhood. Density of unique minimizers then proves equality with the true full-dimensional minimum intervals, as above. $\square$

For use as a vertex certificate, a submitted polygon is additionally required to equal the canonical convex hull of its listed vertices, with positive signed area. This verifies its cyclic order and removes repeated or collinear entries. A test of consecutive turns alone is insufficient, since a self-intersecting polygon can have consistently signed local turns. The one-polygon-per-function requirement is also essential: without it, a genuine minimum cell could be divided into artificial pieces while preserving both dominance and total area. Theorem 4.3, together with the canonical hull check, ensures that the submitted vertices are precisely genuine cell corners.

Groups with only one distinct restricted function have the whole section as their minimum cell and need no separate geometry record. For the remaining source groups, the checker verifies the inventory of group identities. It permits omission of lower-dimensional minimum labels, as justified by Lemma 3.2. On a feasible segment inside a two-dimensional chart, equality of restricted functions is tested by their values at the two endpoints; equality of their ambient coefficient tuples would be too strong.

Theorem 4.5 (Complete ray-list replay). *After independently validating the source model, chart, and section, suppose the checker validates every nontrivial group's coverage certificate as above. It forms the union of section and cell corners, adds intersections of internal edges from different groups, and canonically lifts every resulting point using the independently reconstructed source potentials. Exact equality with the submitted ray list certifies that list as all nonlink standard rays of the supplied sector.*

Proof. The coverage theorems identify the submitted polygons or intervals with the exact source minimum cells, including all cells with nonempty relative interior. Their genuine corners and internal edges consequently form the exact individual group skeletons. The overlay argument in Theorem 3.5 identifies the constructed point set with all vertices of their common refinement; collinear overlaps need only endpoints already present. The lower-dimensional cases are covered by Corollary 3.6. Finally Theorem 3.3 identifies the independently lifted vectors with exactly the required rays. $\square$

4.3 Certificate size and verification complexity

If g groups require geometry records, Lemma 3.4 bounds the number of polygon-vertex incidences by

$$\sum_v (b + 8(m_v - 1)) = gb + 8A.$$

Thus the geometry certificate has $O(k^2)$ vertices when $p \leq 8k$; this charge includes repeated copies of the section boundary. The complete certificate also includes the native ray list, with $O(tk^2)$ scalar entries. Both statements count entries, not their integer bit lengths.

After independent source reconstruction and rank computation, verification of a correct certificate takes $O(k^3 + tk^2)$ rational arithmetic and comparison operations. Indeed, feasibility

and source-boundary scans cost $k \sum_v O(b + m_v) = O(k^3)$. If t_v is the number of native triangle corners in group v , dominance tests cost

$$\sum_v O((b + m_v)t_v) = O(tk),$$

because $\sum_v t_v = 4t$ and $b, m_v = O(k)$. Canonical convex-hull checks and segment intersections fit within the cubic bound, and reconstructing $O(k^2)$ native rays costs $O(tk^2)$, with integer normalization separately charged as above. The cost of reconstructing the native model and proving its rank is additional, not hidden in this geometric replay bound.

For arbitrary malformed certificates, their supplied lengths and integer bit lengths must also be charged. In particular an unrestricted submitted vertex list can require work to parse and reject before its geometric size is known. The displayed parameter bound describes correct certificates, or certificates subject to suitable early size bounds; it does not make parsing an arbitrarily long invalid input free.

5 A sharp abstract quadratic family

The quadratic dependence on the number of potentials is already necessary in a three-dimensional quadrilateral cone. The following family belongs to the algebraic cone-and-potential model. No realization by the matching equations of a triangulated three-manifold is asserted.

Proposition 5.1 (Exact grid count). *For every integer $m \geq 2$, there is a rational model with $\dim K = 3$, two groups of m potentials each, and exactly*

$$p = 2m, \quad R = m^2 + 2m$$

nonlink extreme rays. Its potential coefficients are integers of $O(\log m)$ bits.

Proof. Take $K = \mathbb{R}^3$, $\mathcal{Q} = \mathbb{R}_{\geq 0}^3$, and write $q = (x, y, 1 - x - y)$ on its normalized triangle

$$P = \{(x, y) : x \geq 0, y \geq 0, x + y \leq 1\}.$$

For $j = 0, \dots, m - 1$, define

$$\pi_j^A(q) = j^2 s(q) - 8mj q_1, \quad \pi_j^B(q) = j^2 s(q) - 8mj q_2.$$

On $s(q) = 1$, the first group consists of the functions $j^2 - 8mjx$. Completing the square gives

$$j^2 - 8mjx = (j - 4mx)^2 - 16m^2x^2.$$

The minimizing index is therefore the nearest allowed integer to $4mx$, with ties between adjacent indices at

$$\alpha_j = \frac{2j + 1}{8m}, \quad j = 0, \dots, m - 2.$$

All m minimum cells have nonempty interior. The internal edges are precisely the $m - 1$ vertical segments $x = \alpha_j$ clipped to P . The second group gives the $m - 1$ corresponding horizontal segments $y = \alpha_j$.

Every pair of one vertical and one horizontal segment meets in the interior of P , since $\alpha_i, \alpha_j < 1/4$, so $\alpha_i + \alpha_j < 1/2$. These give $(m - 1)^2$ distinct interior vertices. Each vertical segment has two boundary endpoints, and so does each horizontal segment, giving $4(m - 1)$ boundary vertices. They are distinct from the three triangle corners. Nor can a vertical and

horizontal endpoint coincide on the hypotenuse: that would require $\alpha_i + \alpha_j = 1$. There are no other group vertices or overlay crossings. Hence the complete vertex count is

$$3 + 4(m - 1) + (m - 1)^2 = m^2 + 2m.$$

Theorem 3.3 gives exactly that many nonlink extreme rays. The largest potential coefficient has absolute value $O(m^2)$, proving the bit-length assertion. $\square$

This proves sharp quadratic order for the abstract $O(k + p^2)$ bound, not optimality of the constants in (5). If desired, the model can be padded with identically zero quadrilateral coordinates to width $k = \max(3, m)$; its effective cone and ray count remain unchanged and $p \leq 8k$ then holds. Such padding still establishes no topological realization. Determining which quadratic examples occur in genuine normal matching systems is a separate question. Likewise, a complete polynomial solver for each supplied low-nullity sector gives no bound here on how many sectors a global recognition procedure must inspect.

6 A realized three-dimensional family with exact work counts

The following family separates the geometric advantage of minimum cells from questions about which sectors an unknot recognizer must visit. Every member is an actual triangulated solid torus. Its supplied sector has three-dimensional quadrilateral matching space, seven nonlink standard rays, and exponentially large primitive coordinates. All statements quantified over the family are proved below; the finite computational checks are reported separately at the end.

6.1 Face pairings and the underlying manifold

Number tetrahedron vertices 0, 1, 2, 3, and number a face by its opposite vertex. A permutation written (p_0, p_1, p_2, p_3) sends vertex j to p_j . A normal row is ordered

$$(t_0, t_1, t_2, t_3; q_0, q_1, q_2),$$

where quadrilateral types 0, 1, 2 avoid, respectively, the opposite edge pairs (01, 23), (02, 13), and (03, 12). These are the conventions of the native fixture.

For $n \geq 1$, let T_n contain tetrahedra $0, \dots, n - 1$, with face pairings

$$(i, 0) \xrightarrow{(2,1,3,0)} (i + 1, 2), \quad (i, 1) \xrightarrow{(0,3,1,2)} (i + 1, 3) \quad (0 \leq i < n - 1), \quad (7)$$

$$(n - 1, 0) \xrightarrow{(1,2,3,0)} (n - 1, 1). \quad (8)$$

The inverse pairings are understood. Its unpaired faces are (0, 2) and (0, 3). Form M_n by adding tetrahedra $n, n + 1$ and pairing

$$(0, 2) \xrightarrow{\text{id}} (n, 2), \quad (0, 3) \xrightarrow{\text{id}} (n + 1, 3). \quad (9)$$

Thus $t = n + 2$. Allow type 2 in each original tetrahedron and type 1 in each cap, so $k = t$.

The face closures in this description need not embed: all vertices of T_n are identified. The following observation handles that issue without treating a quotient face as an embedded closed disc.

Lemma 6.1 (Boundary caps with identified outer boundaries). *Let $S = \partial M$ be the boundary of a compact PL manifold. Suppose an abstract PL disc D is attached by a characteristic map $f : D \rightarrow S$ that embeds its interior, with $f(\partial D) \cap f(\text{int } D) = \emptyset$. Identifications between points of ∂D are permitted. Attach a PL ball along a disc in its boundary identified with D . The resulting space is PL homeomorphic to M . Its boundary replaces D by the complementary abstract boundary disc, with the same outer boundary identifications.*

Proof. Choose a PL function $\rho : D \rightarrow [0, 1]$, positive in the interior and zero on the boundary, after subdividing D if necessary. The region

$$B_\rho = \{(p, s) : p \in D, 0 \leq s \leq \rho(p)\}$$

is a PL ball whose lower and upper discs meet along ∂D . A boundary homeomorphism from the given ball to this model, fixed on the attaching disc, extends across the balls by coning a common boundary subdivision. Since every identification under f occurs where $\rho = 0$, the height function descends to a continuous PL function $\bar{\rho}$ on S , extended by zero off $f(D)$. The pushout is therefore the original manifold with the region $0 \leq s \leq \bar{\rho}(q)$ added to an exterior boundary collar.

Take an interior collar $S \times [-1, 0]$. The enlarged collar has vertical intervals $[-1, \bar{\rho}(q)]$. Triangulate S so that $\bar{\rho}$ is affine on each simplex. Corresponding triangulations of the prisms between the lower and upper graphs give a PL homeomorphism to $S \times [-1, 0]$, fixing the lower boundary. Extend it by the identity outside the collar. This also proves the assertion at identified vertices and edges: it is an ambient collar construction, not merely a comparison of the two open faces. $\square$

Proposition 6.2 (Realization as solid tori). *The space T_n is an orientable solid torus with one vertex. In its front boundary tetrahedron its three distinct boundary-edge classes are*

$$a_n = [01], \quad b_n = [02] = [13], \quad c_n = [03] = [12].$$

The space M_n is an orientable solid torus with three boundary vertices.

Proof. For T_1 , the edge classes are $\{01\}$, $\{02, 13\}$, and $\{03, 12, 23\}$, all boundary edges. Its four vertex-link triangles are joined in the chain $1, 2, 3, 0$, giving one disc link; its edge links are intervals, without reversed self-identifications. The face pairing is orientation compatible. Thus this is a compact orientable manifold. Removing collars of its boundary vertices, edges and faces leaves a ball around the tetrahedron centre and one handle through the paired faces. There are no interior edge or vertex dual blocks. Consequently the manifold is a regular neighbourhood of a one-loop graph, hence an orientable solid torus.

To pass from T_m to T_{m+1} , put a new front tetrahedron before the old ones, using (7). Its attaching disc is the union of faces $0, 1$, with internal diagonal 23 . Both occurrences of this diagonal map to c_m ; the four outer edges map to a_m, b_m, a_m, b_m . These three edge classes are distinct. The interior of the abstract attaching disc therefore embeds in the old boundary: its only boundary identifications concern the outer edges and vertices. Lemma 6.1 applies. The new boundary classes are $a_{m+1} = [01]$, $b_{m+1} = a_m$, and $c_{m+1} = b_m$; the old c_m becomes interior. The new vertices join the unique old vertex. This proves the induction and the stated distinctness.

Each attachment in (9) satisfies the one-face case of the same lemma. The remaining original face is unaffected by the first attachment. Each cap contributes its opposite vertex as a new boundary vertex. In link language, the three old corner intervals are distinct intervals on the old vertex-link boundary; the cap adds triangular flaps there, while its new vertex has a disc link. These observations also explain directly why the quotient face closures cause no singularity. $\square$

6.2 The complete matching cone and its canonical section

Set $F_0 = 0$, $F_1 = 1$, and $F_{j+2} = F_{j+1} + F_j$, and abbreviate $A = F_{n+1}$, $B = F_{n+2}$.

Theorem 6.3 (Full coordinates, including vertex links). *Every real standard matching solution in the prescribed support has the following unique form, with parameters x, y, z, b, u, v :*

$$\text{row } i : (F_{n-i+1}x + b, F_{n-i+1}x + b, b, b; 0, 0, F_{n-i}x), \quad 0 \leq i < n, \quad (10)$$

$$\text{row } n : (Ax + b - y, Bx + b, u, b; 0, y, 0), \quad (11)$$

$$\text{row } n + 1 : (Bx + b, Ax + b - z, b, v; 0, z, 0). \quad (12)$$

It is nonnegative precisely when

$$x, b, u, v \geq 0, \quad 0 \leq y, z \leq Ax + b. \quad (13)$$

Its quadrilateral matching space has dimension three; its quadrilateral cone is parametrized by $(x, y, z) \in \mathbb{R}_{\geq 0}^3$. For such a triple put

$$h = \max(0, y - Ax, z - Ax). \quad (14)$$

The canonical normal vector has $b = h$ and $u = v = 0$. Every nonnegative solution uniquely adds $(b - h)$ copies of the old vertex link and u, v copies of the two new vertex links to that canonical vector.

Proof. Substitute type 2 into the base face equations, working backwards from (8). The triangle entries are (X_i, X_i, Y_i, Y_i) , with quadrilateral entry W_i , and the equations reduce to

$$\begin{aligned} X_i &= X_{i+1} + W_{i+1}, & Y_i &= Y_{i+1}, & Y_i + W_i &= X_{i+1} \quad (0 \leq i < n - 1), \\ X_{n-1} &= Y_{n-1} + W_{n-1}. \end{aligned}$$

Taking $x = W_{n-1}$ and $b = Y_{n-1}$ gives $W_i = F_{n-i}x$, $X_i = F_{n-i+1}x + b$, and $Y_i = b$. Conversely these formulas satisfy every base equation. Writing the first cap triangles as r_j , its attaching equations are $r_0 + y = Ax + b$, $r_1 = Bx + b$, and $r_3 = b$; $r_2 = u$ is free. For the second cap they are $r_0 = Bx + b$, $r_1 + z = Ax + b$, and $r_2 = b$; $r_3 = v$ is free. This proves the complete linear parametrization and (13).

Every nonnegative (x, y, z) has a lift by choosing $b \geq h$. The last base quadrilateral and the two cap quadrilaterals recover these three parameters, so the parametrization is injective and its matching dimension is three. At the old global vertex the minimum triangle entry is

$$\min(b, Ax + b - y, Ax + b - z) = b - h;$$

all other old-vertex entries are at least b . The two new minima are u, v . Subtracting these three minima proves the canonical and link assertions, in the standard sense of canonical quadrilateral-to-standard lifting [1]. $\square$

Proposition 6.4 (Seven nonlink rays and three active cells). *The complete standard sector has ten primitive rays: its three vertex links and the canonical lifts of*

$$\begin{aligned} (1, 0, 0), (1, A, 0), (1, 0, A), (1, A, A), \\ (0, 1, 0), (0, 0, 1), (0, 1, 1). \end{aligned} \quad (15)$$

The normalized quadrilateral section has three minimum cells, three internal edges and seven vertices. The one-group output bound is attained: $3 + 2(3 - 1) = 7$.

Proof. First discard the independent coordinates u, v . Normalize the remaining cone by $Ax+b=1$. Equation (13) becomes the cube

$$(Ax, y, z) \in [0, 1]^3, \quad b = 1 - Ax.$$

Its eight vertices give four rays with $x=1, b=0$ and $y, z \in \{0, A\}$, and four with $x=0, b=1$ and $y, z \in \{0, 1\}$. The latter corner $y=z=0$ is the old vertex link; the other three are the last three rays in (15). Restoring u, v adds the two other link rays. The displayed nonlink vectors are primitive because either the last base quadrilateral or a cap quadrilateral is one.

In the base-offset gauge the only active minimum functions are $0, Ax-y, Ax-z$. Their three regions are

$$y, z \leq Ax; \quad y \geq Ax, y \geq z; \quad z \geq Ax, z \geq y.$$

The actual normalization is $(B-1)x + y + z = 1$, since $\sum_{j=1}^n F_j = B-1$. On this triangle the three internal edges meet at the direction $(1, A, A)$ and end at $(1, A, 0)$, $(1, 0, A)$, and $(0, 1, 1)$. Together with the original triangle vertices these are exactly seven vertices, proving the geometric assertions. $\square$

6.3 A complete disc census for every integral parameter

Lemma 6.5 (The base meridian). *The base vector (10) with $x=1, b=0$ is one essential meridian disc D_n .*

Proof. Any normal component has nonnegative integral base parameters. A decomposition of this vector has total $x=1, b=0$, so exactly one component can be nonzero. This proves connectedness without expanding the Fibonacci coordinates.

Here is an explicit Euler-characteristic check. Write $C = F_{n+3} = A + B$. The three boundary-edge intersection numbers, in the order a_n, b_n, c_n , are C, B, A . The interior edges inherited from T_m , $1 \leq m < n$, have weights F_{m+1} . Thus the normal cell decomposition has

$$V = 2C + B - 2, \quad E = 4C + 2B - 8, \quad F = 2C + B - 5.$$

Indeed there are $2(C-2)$ triangles and $B-1$ quadrilaterals; their sides, with boundary sides counted once, give the displayed edge count. The vertex count follows by summing the edge weights and using $\sum_{m=1}^{n-1} F_{m+1} = B-2$. Hence $V - E + F = 1$. The connected surface has nonempty boundary and Euler characteristic one, so it is a disc.

An inessential circle on the boundary torus has even mod-two intersection with every closed edge loop. Consecutive Fibonacci numbers A, B are coprime, so they cannot both be even. The listed boundary intersections therefore show that ∂D_n is essential. In a solid torus an essential properly embedded disc is a meridian disc. $\square$

Theorem 6.6 (Integral component classification). *For nonnegative integral (x, y, z) , the canonical vector consists of x essential meridian discs and h inessential discs, where h is given by (14). Its component count and Euler characteristic are both $x+h$. More generally, the full integral solution (10)–(12) consists of x meridian discs and $b+u+v$ inessential discs.*

Proof. First take $u=v=0$. The base restriction is the disjoint union of x parallel copies of D_n and b small vertex-link discs. These can be chosen disjoint: the meridian avoids the vertex, and its two-sided regular neighbourhood supplies the parallel copies. Their normal coordinates are exactly the base rows. The usual uniqueness of an embedded normal surface with specified

coordinates identifies this union with the base restriction: stack the prescribed pieces in each tetrahedron and match the ordered normal arcs across its faces.

Every cap piece present when $u = v = 0$ meets its attaching face in exactly one arc. This holds for each quadrilateral and for each triangle whose vertex belongs to that face. The sole triangle type missing the face is its opposite-vertex triangle, and that coordinate is zero. Because the three face edges represent distinct global edges, these arcs are embedded boundary intervals, mutually disjoint within the attaching face, even though its vertices coincide in the quotient.

Each cap disc therefore attaches to one base disc along one boundary interval. Such an attachment leaves a disc and cannot join two components. All cap pieces attach, so none creates an additional component. Apply the argument successively to the two caps; the second original face remains available after the first attachment. This gives exactly $x + b$ discs.

Essentiality also survives these attachments. In the abstract attaching disc, each old normal arc is replaced on the complementary boundary disc by the remainder of the same cap disc's boundary. The replacement arcs are disjoint and have exactly the same endpoints and pairing. Two disjoint arc systems in discs with the same boundary pairing are related by a PL homeomorphism fixed on the boundary: cut successively along outermost arcs and extend over the resulting discs. This homeomorphism respects every quotient identification on the outer boundary and hence descends to a homeomorphism of the boundary tori, identity off the replaced face. It carries old boundary curves to new ones. Thus meridian boundaries remain essential and link boundaries remain inessential. This argument uses abstract discs and their identified boundaries, rather than assuming embedded face closures.

The triangles counted by u, v are disjoint links of the two new boundary vertices. Adding them contributes $u + v$ inessential discs. Finally put $b = h$, $u = v = 0$ for the canonical statement. $\square$

Corollary 6.7 (Essential standard rays beyond Q-rays). *All seven nonlink rays are discs. The four with $x = 1$ are essential; the three with $x = 0$, including $(0, 1, 1)$, are inessential. Only $(1, 0, 0)$, $(0, 1, 0)$, and $(0, 0, 1)$ are quadrilateral extreme directions. Consequently three of the four essential standard rays are absent from the Q-extreme-ray list.*

6.4 Exact work performed by the full equality arrangement

The counts below concern the standard phase of `sector_rays`, using the `arrangement` method without a work cap. It considers every pair of distinct projected hyperplanes, deduplicates nonnegative primitive directions, lifts each retained direction, and tests standard extremality. These counts describe work after kernel construction.

Proposition 6.8 (Attempted, positive and rejected directions). *For this family the full equality arrangement has*

$$H = 2n + 6, \quad \text{pair attempts} = \binom{2n + 6}{2}, \quad \text{distinct nonnegative directions} = (n + 1)^2 + 3. \quad (16)$$

Exactly $(n - 1)(n + 3)$ of these directions are rejected by the standard-extremality test; seven are emitted.

Proof. The distinct old-vertex potentials in the base-offset gauge are

$$\{cx : c \in C_n\} \cup \{Ax - y, Ax - z\}, \quad C_n = \{0, F_2, F_3, \dots, F_{n+2}\}.$$

The set C_n has $n + 2$ elements, because the Fibonacci sequence is strictly increasing from F_2 onwards. Pairwise equalities and quadrilateral coordinate hyperplanes give precisely

$$x = 0, \quad y = (A - c)x \ (c \in C_n), \quad z = (A - c)x \ (c \in C_n), \quad y = z. \quad (17)$$

The planes $y = 0, z = 0$ already occur at $c = A$. All displayed planes are distinct, proving $H = 2n + 6$. Every pair is independent in three dimensions, and the explicit loop visits every pair.

For $x > 0$, normalize $x = 1$. Put

$$S_n = \{A - c : c \in C_n, \ c \leq A\}.$$

Only B is excluded by $c \leq A$, so $|S_n| = n + 1$. The intersections in the nonnegative quadrant are exactly the grid $(y, z) \in S_n \times S_n$: intersections with $y = z$ merely repeat its diagonal entries. The remaining $x = 0$ directions are $(0, 1, 0)$, $(0, 0, 1)$, and $(0, 1, 1)$. All grid directions are integral and primitive, since the last base quadrilateral is one. This proves the positive-direction count after deduplication.

Every grid point satisfies $0 \leq y, z \leq A$, hence $h = 0$. Among them only the four corners of this square are standard extreme; every other point is a nontrivial convex combination of corners on the linear canonical-lift cell. The three $x = 0$ directions are all extreme. The rejected count is therefore $(n + 1)^2 - 4 = (n - 1)(n + 3)$. In fact every rejected grid direction is itself a connected meridian disc, by Theorem 6.6; its rejection concerns extremality rather than topology. $\square$

The minimum construction retains three cells and seven directions for all n , whereas the equality arrangement performs a growing quadratic number of distinct positive lifts and extremality checks. The additional $n + 1$ inactive potentials are nonnegative multiples of x , dominated by the zero potential. Source construction, projection costs, coordinate arithmetic and serialization are separate from the exact loop counts in (16).

6.5 Encoding size and finite independent evidence

Proposition 6.9 (Input and output bit lengths). *With an explicit numbered face-pairing table, the family input has $\Theta(n \log(n + 1))$ bits. The seven complete primitive normal vectors have $\Theta(n^2)$ bits in total. Their largest coordinate is $B = F_{n+2}$. A specialized implementation of the displayed formulas can construct and serialize the seven vectors in $\Theta(n^2)$ bit operations using ordinary binary addition.*

Proof. There are linearly many tetrahedron references, a linear fraction of which require $\Theta(\log(n + 1))$ bits; all permutations have constant size. Each output contains linearly many coordinates, each with $O(n)$ bits. Conversely the base coordinates of a meridian contain $F_1, \dots, F_n$, whose binary lengths sum to $\Theta(n^2)$; for example a linear fraction of these entries have linear bit length. The cap coordinate B realizes the maximum. Generate the Fibonacci list by successive additions and substitute into the seven formulas. The additions and the full output require $O(n^2)$ bit work, matching the output-size lower bound. $\square$

A succinct family recipe needs only the integer n , but the native solver receives the expanded triangulation. The specialized formula bound is not a bound on all stages of the generic sector implementation. Likewise exponentially many normal pieces need not be expanded to classify their components; compressed orbit methods provide the relevant general setting [7].

The retained `family_audit.py` checks $n = 1, 2, 4, 8$. It verifies 300 prescribed parameter-grid instances against the native canonical lift, regenerates the seven rays, and independently replays four complete coverage certificates. Native compressed component censuses and their independent checker classify all 28 ray instances and twelve additional mixed directions. Regina 7.4 independently classifies all 28 rays and verifies that each source is a solid torus. Its complete standard vertex lists contain respectively 35, 39, 85, 375 surfaces; restricting each list to the supplied sector and removing pure links gives exactly the seven proved rays. These finite results test the implementation against the formulas and an external oracle; the all-size conclusions follow from the preceding proofs.

7 Implementation and the meaning of its certificates

The proposed change adds three runtime modules and keeps the maintained dispatcher unchanged. `sector_sparse.py` compiles and reuses the source. `sector_planar.py` constructs the minimum diagrams, enumerates their vertices, emits full primitive rays, and optionally composes them with the existing disc-component checker. `sector_planar_verify.py` checks coverage from the original native matching system.

7.1 Exact geometry and output

The producer computes a rational basis of the selected quadrilateral kernel. A normalized affine chart uses actual quadrilateral coordinates as its independent variables. Since every feasible coordinate lies in $[0, 1]$, the initial unit interval or square is justified; it is not a numerical bounding-box guess. The source nonnegativity inequalities are then intersected exactly. A zero sum functional on the kernel correctly implies that the nonnegative cone has no nonzero point.

For a polygon, the producer projects every retained potential into the two-dimensional chart once. It clips a candidate cell by the dominance halfplanes and keeps only positive-area cells. The representation removes consecutive collinear vertices, fixes counterclockwise order, and rotates to a deterministic first corner. For a feasible segment, it identifies forms by their values at the two endpoints and constructs a sorted lower envelope of lines. Thus forms that differ outside the feasible affine hull are correctly treated as identical when necessary.

The producer evaluates all quadrilateral and projected triangle forms at each overlay vertex, normalizes quadrilaterals to their primitive integer direction, subtracts the group minima, expands contracted classes, and validates the resulting native coordinates. All geometry uses Python rational arithmetic. No floating-point predicate determines feasibility, equality, extremality, or output identity. Floating point occurs only in wall-clock timing and in the plotted presentation of exact formulas.

7.2 A checker that does not rerun the producer

The checker does not import the planar producer or the sparse constructor. It reconstructs triangle-difference potentials from the native face matching rows before contraction and computes the quadrilateral consistency kernel independently. Its affine-chart test combines matching, normalization, an actual-coordinate left inverse, and an independent dimension computation. A submitted polygon is checked by an independent convex hull and source-supported facet test. Degenerate or empty sections are checked by independent line-intersection feasibility.

For each group, the checker verifies the label inventory, nondegenerate convex cell geometry, dominance at every cell vertex, distinct affine-function labels, and exact total area or length.

It reconstructs overlay vertices with line-equation intersection formulas, whereas the producer uses a vector interpolation formula. Finally it independently lifts the complete point set and compares the entire ordered list of native vectors with the certificate. The coverage theorem justifies this comparison as a completeness claim, including a genuinely empty sector.

This is algorithmic independence of the important reduction and geometry paths, not a claim of a formally verified trusted kernel. Both sides necessarily use the same source format, integer codec, and maintained normal-coordinate validation contract. The checker also reuses an existing independent linear-algebra helper. Differential tests against the old arrangement and Regina provide further independent evidence; none substitutes for the mathematical proof.

The schema `normal-sector-planar-v1` records the source digest, allowed types, raw matching dimension, feasible section dimension, affine chart, section vertices, labelled group cells, and complete ray list. Rational coordinates are reduced numerator–denominator pairs. Large integers use the maintained integer codec, including hexadecimal representations beyond its threshold; replay must use the same semantics rather than assume every integer is a decimal JSON number. The tests include a 20,000-bit value.

7.3 Composing with a disc search

The function `discover_in_planar_sector` first obtains complete coverage, then asks the maintained compressed component routine about each ray. Each component-count certificate is independently replayed before it is used. It does not skip a ray merely because its total Euler characteristic is nonpositive; that would require an additional component argument. The composition returns one of three semantically distinct outcomes:

DISC_FOUND

A positive essential-disc witness for the supplied triangulation, using the existing `normal-sector-witness-v1` schema and its maintained independent verifier. Recognition of an input knot still requires its certified exterior correspondence. **NO_VERTEX_DISC_IN_SECTOR**

A full coverage certificate plus a zero essential-disc-component certificate for every enumerated nonlink standard ray. The new `normal-sector-planar-disc-scan-v1` verifier replays both parts. This excludes such components in these vertex rays of this supplied sector. It is not a global negative knot verdict. **INCONCLUSIVE**

A requested resource allowance was exhausted. The result contains no certificate of complete exhaustion.

A matching nullity above three is an explicit unsupported-input error for this specialized routine. The ray generator raises the maintained search-limit exception when its emission allowance is exhausted; the certifying API returns an inconclusive result instead of packaging a partial list as complete. Cooperative cancellation callbacks propagate their exceptions, including when the callable object itself has false Boolean value. User-supplied callbacks are therefore tested by their call result, not by truthiness.

This composition makes the local research result usable by the recognizer without silently modifying its global search policy. It also exposes the distinction between a cheap positive witness and complete negative exhaustion: on the growing family, the old arrangement can emit all actual rays before spending much more time ruling out the remaining candidate directions.

7.4 A minimal native example

From the included code directory:

```
from planar_sector_research.fixtures import double_capped_fibonacci
from fastunknot.sector_sparse import PreparedSectorSource
from fastunknot.sector_planar import certify_planar_sector
from fastunknot.sector_planar_verify import verify_planar_sector_certificate

fixture = double_capped_fibonacci(8)
tri = fixture["triangulation"]
source = PreparedSectorSource(tri)
result = certify_planar_sector(
    tri, fixture["allowed_types"], source=source)
assert result["status"] == "COMPLETE"
assert len(result["rays"]) == 7
assert verify_planar_sector_certificate(tri, result["certificate"])
```

The source may now be reused for other supports on the same triangulation, subject to its read-only contract. The example requires no Regina installation. Regina is an optional independent oracle for the full experimental reproduction.

8 Exact audits and controlled performance measurements

The experiments answer three separate questions: whether the new algorithm emits the complete correct ray set, whether its certificates can be replayed independently, and whether the implementation improves the intended stage against a relevant incumbent. Exact output comparisons use primitive full standard-coordinate vectors, not only ray counts or quadrilateral supports. Timings are secondary to these correctness checks and do not establish an asymptotic exponent.

8.1 Frozen complete oracle and unchanged support selection

The principal audit reuses the preceding report’s frozen discovery corpus: 48 valid compact orientable triangulations with one torus boundary, including solid tori, knot exteriors, and connected-sum controls. Its SHA-256 is

```
0433e6c7d99bb402b79909467a7eada862ca
004dc4082ee25794add0e712914e
```

The driver preserves the prior support-selection rule: occupied supports of the observed quadrilateral and standard rays, the empty support, all supports on one or two tetrahedra, all compatible supports for sources with at most three tetrahedra, and 64 deterministically seeded full sectors for larger sources. Duplicate supports are removed. This gives exactly the same 9,995 selected source/support pairs as before. It is a finite, oracle-informed audit design, not an unbiased input distribution or a proposed complete sector-discovery algorithm.

One prepared source is built per triangulation. Every eligible sector is enumerated completely. Its output is compared with the frozen complete standard-coordinate vertex list restricted to that sector, after pure links are removed. The resulting figures are shown in Table 1. The new method covers all 9,344 previously eligible cases and 463 additional cases of raw nullity three. The 188 higher-nullity cases are retained as out of scope.

All 9,807 eligible ray sets agree exactly. The non-Q counts show why a quadrilateral-vertex oracle alone would not test the claim being made. The distinct-source/ray figures remove

Table 1: Complete frozen-corpus audit. An occurrence counts a ray again if it lies in several selected sectors.

Quantity	All eligible $d \leq 3$	New $d = 3$ cases
Sectors	9,807	463
Standard-ray occurrences	7,674	2,821
Occurrences absent from Q-vertex lists	2,279	1,475
Essential-disc occurrences	332	129
Non-Q essential-disc occurrences	118	84
Distinct source/ray pairs	1,663	969
Distinct non-Q source/ray pairs	1,046	759
Distinct essential source/ray pairs	105	55
Distinct non-Q essential source/ray pairs	57	47

multiplicity from repeated appearances in overlapping sectors. They should not be added to the occurrence counts.

Table 2: Raw matching nullity and actual projective dimension. The degenerate cases occur in genuine triangulations.

Raw d	Empty	Point	Segment	Polygon	Total
0	5,357	0	0	0	5,357
1	890	1,947	0	0	2,837
2	29	140	981	0	1,150
3	1	3	39	420	463

The dimension breakdown in Table 2 is important. It would be incorrect to treat every raw three-dimensional kernel as a positive-area polygon or to infer emptiness from a missing two-dimensional interior. Thirteen retained coverage certificates span every observed pair of raw and feasible dimension. A separate replay-only process accepts all thirteen. This does not mean every corpus certificate was serialized and replayed: the full audit compares every output, while independent retained replay covers the selected dimension strata.

Eight complete standard-coordinate lists were freshly regenerated using Regina 7.4 and matched the frozen lists: the four cap controls with identifiers `cap_1_2_2`, `cap_1_2_3`, `cap_3_5_2`, `cap_3_5_3`; finite trefoil and figure-eight interiors; `solid_torus_sum_s2xs1`; and `fibonacci_1st_10`. Their complete lists contain 35, 83, 52, 120, 55, 446, 69, and 107 standard vertices, respectively. The source snapshot remained unchanged throughout the audit. The digest of all eligible computed output sets is

3ebe4355456c91c036c5ef40b0c81361b17
aa92666e87a1130195421f3fa1687.

The largest observed sector output has seven rays, the largest number of nontrivial active groups is two, and the largest observed number of cross-group segment-pair tests is one. Thus this corpus is useful for correctness and degeneracies, but it does not empirically realize the growing quadratic worst case.

8.2 Additional independent and adversarial checks

An additional differential driver compares 1,646 genuine eligible sectors, with 2,712 ray occurrences, against the incumbent arrangement and an independent active-rank test. It also checks 240

abstract potential cones, containing 449 output vertices, against an independently generated all-equalities arrangement followed by extremality filtering. These audits are supplementary; their sources overlap earlier controls and their counts should not be summed into an asserted number of distinct triangulations.

The focused suite contains 31 passing tests across the sparse constructor, exact planar geometry, adversarial coverage replay, and disc-search composition. The unchanged maintained sector-integration suite adds eight passing tests. The focused checks include all 64 allowed/absent type choices on a small two-cap source; 292 exact sparse/incumbent kernel-field comparisons; 180 deterministic random rational clipping cases against an independent halfplane-intersection oracle; abstract grids; inactive equalities; coincident minimum edges; and raw nullity-three cones with empty, point, and segment sections.

Coverage mutations exercise wrong sources, charts, labels, missing and duplicated cells, reversed polygon order, shrunk sections, missing or reordered rays, malformed rationals, Boolean values in integer fields, and cancellation callbacks. The checker is also exercised with the producer disabled. The discovery tests replay a positive witness, replay empty and nonempty sector exhaustion, reject incomplete per-ray disc checks, and ensure a ray allowance cannot manufacture an exhaustion certificate. These are targeted regression tests for material correctness risks, not a claim of exhaustive testing of every malformed input to all inherited repository modules.

8.3 Geometric family verification

The double-cap family was checked at $n = 1, 2, 4, 8$, independently of the timing experiment. All 28 predicted ray vectors are classified by both native compressed component certificates with replay and Regina: each is a connected orientable disc with one boundary component; four per source are essential and three are inessential. The audit additionally checks 300 deterministic canonical-formula grid points and twelve mixed directions, confirming the predicted counts $x + h$ components and x essential discs.

Fresh complete Regina standard lists at these four sizes contain 35, 39, 85, and 375 vertices. Restricting each complete list to the designated sector leaves exactly seven nonlink rays. Four further complete coverage certificates are replayed in a separate process. The all- n assertions in Section 6 rest on the face-equation and topological proofs; these finite calculations check the fixture conventions and implementation against those proofs.

8.4 Paired complete-enumeration timings

The primary performance experiment uses $n = 1, 2, 4, 8, 16, 32$ base tetrahedra and $t = n + 2$ total tetrahedra. For each size it retains a warmup and three paired rounds, alternating the execution order. The *full* incumbent arm builds the original kernel and enumerates with the maintained automatic dispatcher; the full new arm builds the sparse kernel and enumerates with the planar routine. The dispatcher selects the old arrangement on this family. Two additional arms use the *same already constructed* incumbent kernel, isolating enumeration after matching.

The timed task is complete ray-list exhaustion. Full arms include native geometry preparation, exact matching construction, enumeration, and all native coordinate checks performed during lifting. Fixture generation, shared-kernel setup, hashes, formula-vector comparison, and serialization are outside the timers. Disc-component classification, independent coverage replay, sector discovery, and whole-knot recognition are not timed here. Every completed arm returns exactly the seven formula rays; the old hyperplane, positive-direction, rejected-direction, and pair-candidate counters match the all-size formulas.

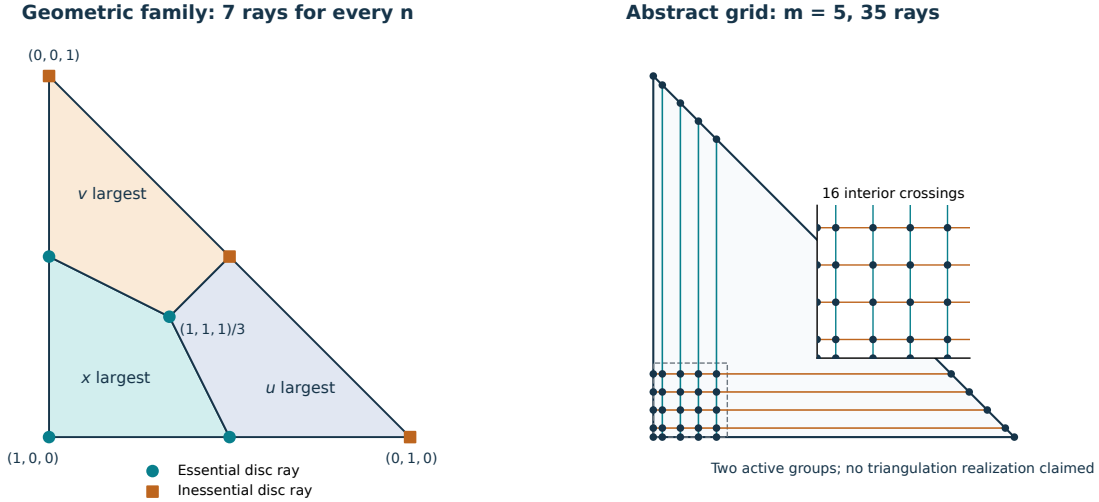


Figure 1: Exact geometric models. Left: the seven-vertex two-cap diagram, drawn in the rescaled parameter section $x + u + v = 1$ with $u = y/A$, $v = z/A$. Each region corresponds to the largest of x, u, v , or equivalently the active minimum after a common affine shift. Right: the abstract two-group grid at $m = 5$, with $m^2 + 2m = 35$ vertices. Only the left diagram is asserted here to come from the exhibited triangulated solid-torus family.

n	Full construction and enumeration			Enumeration on the same kernel		
	Incumbent	Planar	Paired ratio	Incumbent	Planar	Paired ratio
1	4.238	2.720	1.47	3.002	2.363	1.27
2	8.550	3.057	2.83	8.224	2.898	2.84
4	46.490	5.209	8.90	47.096	4.367	10.79
8	537.411	8.730	61.64	530.946	6.916	73.80
16	8 572.669	19.861	482.82	8 749.966	11.624	758.84
32	40 000 cap	48.458	—	40 000 cap	25.997	—

Table 3: Complete seven-ray enumeration for the double-capped Fibonacci family. Times are medians of three alternating-order paired rounds, in milliseconds; one warmup per arm is retained in the raw data and excluded from these medians. Paired ratios are medians of within-round incumbent/planar time ratios, not ratios of the displayed medians. For example, at $n = 16$ the full paired ratio is 482.82, whereas the ratio of medians is 431.64. The two post-kernel arms receive exactly the same immutable incumbent-built kernel. At $n = 32$, all eight incumbent warmup/round runs reach the cooperative 40 000 ms cap after emitting seven rays but before proving enumeration complete; the cap entries are allowances, not completed-run medians, and no speed ratio is reported.

n	Prior envelope (ms)	Planar (ms)	Paired prior/planar	Ratio of medians
8		2.193	2.415	0.92
32		8.534	8.741	0.97

Table 4: Control against the earlier optimized nullity-two envelope enumerator, loaded unmodified from the preceding research package. Both arms use the same incumbent-built kernel and return the same three formula rays. Times are medians of three paired rounds; values below one in either ratio column indicate a slower planar implementation. These controls show no measured advantage over the already optimized nullity-two method.

k	Complete construction (ms)			Paired incumbent/sparse ratios	
	Incumbent	Sparse once	Sparse reused	Reused source	Both cached
1	72.133	81.463	8.004	8.84	1.88
8	44.435	51.755	5.638	7.88	2.06
32	61.126	64.297	12.041	4.74	2.02

Table 5: Separate sparse-constructor measurements on supplied sparse sectors of $t = 1024$ layered solid tori. Times are medians of five paired rounds and include unchanged exact rational elimination. The reused-source column excludes its one-time preparation, while both ordinary single-call constructors include preparation. The last column uses an additional control where both constructors receive already validated geometry; it isolates sparse kernel construction from validation reuse. Ratios are medians of within-round ratios. The single-call sparse constructor is slower on these cases. These are construction measurements, separate from complete ray enumeration and from whole-knot recognition. All nine constructor cases and every raw sample are retained in the data files.

The $n = 32$ incumbent hits the cooperative 40-second completion limit in both warmups and all six timed incumbent rounds. It has already emitted seven genuine rays, but has not finished eliminating the other candidate directions. Its records are resource-limited, and no speedup ratio is assigned to that size. This is precisely why the benchmark does not measure merely the time to observe seven answers or the time to the first positive disc. A cooperative cap can overshoot by the duration of an indivisible exact operation.

The rank-two control loads the preceding optimized `sector_envelope.py` unchanged and compares it with the new method on the same kernel. The older specialized method is slightly faster in these measurements. This negative result argues for preserving the earlier branch when designing an eventual dispatcher. It would be misleading to claim improvement over that method by comparing only with an older all-equalities implementation.

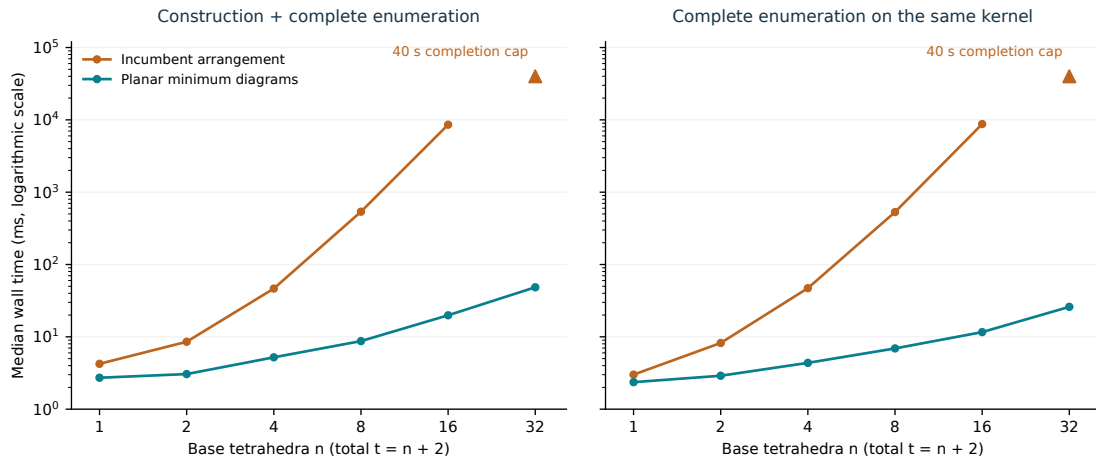


Figure 2: Median complete-enumeration wall time over three paired rounds, with logarithmic axes. The incumbent $n = 32$ value is a resource limit, displayed as a cap marker rather than a completed timing. These are stage measurements on one growing family, not a fitted worst-case exponent or a whole-recognizer speedup.

8.5 Construction-only reuse controls

The separate sparse-constructor experiment uses nine supplied-support cases on growing layered solid tori and five rounds per case. It distinguishes original complete construction, one-shot sparse construction, reused sparse construction, and a control in which both constructors receive reusable prepared geometry. Using ratios of median times across these cases, reused sparse construction is 1.38–9.01 times as fast as original complete construction. When both reuse geometry, the measured ratio is 1.10–2.14. One-shot sparse construction gives ratios 0.76–0.99 and is slower in this sample.

At $t = 1024, k = 32$, original complete construction has median 61.126 ms, one-shot sparse construction 64.297 ms, and reused sparse construction 12.041 ms. The matched prepared-geometry control is 27.334 ms for the old builder and 12.802 ms for the sparse builder. Dense label entries fall from 196,512 to 4,032. This gives a concrete allocation explanation while exposing the cost of preparing a source that will be used only once.

All reported timings are observations from Python 3.12.14 in the supplied execution environment. Exact source digests, raw wall and CPU times were recorded, execution order, counters, caps, and output hashes accompany the package. Three or five rounds on selected sources do not support confidence claims about unrelated hardware or a general input distribution.

9 What is still required for quasi-polynomial recognition

The new theorem removes a local obstacle: a supplied sector of raw matching dimension three has polynomially many relevant standard rays, with a polynomial exact enumeration and coverage proof. It does not bound how many sectors are required. Even listing all maximal quadrilateral choices gives 3^t possible sectors, each containing its own coordinate faces. Polynomial work per selected sector therefore remains compatible with exponential total work. The fixed corpus demonstrates the utility of the new eligibility class, not its completeness for all unknots.

The following conditional statement records exactly what the local result can contribute. It is the dimension-three extension of the preceding report’s complete-family accounting statement [10]; its point is to expose hypotheses, not to claim they have been established.

Corollary 9.1 (Conditional composition with a complete search). *Suppose a deterministic procedure on an n -crossing knot diagram visits at most $S(n)$ source-bound states and supplied sectors, with the following properties:*

1. *all encoded state sizes, matching systems, and normal-vector bit lengths are polynomial in n ;*
2. *each sector has matching nullity at most three, or is reduced to one by a checked certificate preserving all relevant candidates;*
3. *every state construction, failed branch, restart, transition, terminal decision, and certificate replay together costs $S(n) \text{ poly}(n)$;*
4. *the necessary essential-disc-component predicates can be produced and independently replayed in polynomial bit time for each emitted vector;*
5. *the certified source correspondence and selection theorem guarantee that every unknot yields a relevant disc in at least one enumerated ray, while the terminal rules justify the other outcome.*

Then the local enumeration, coverage replay, and disc tests can be incorporated into a complete recognizer in $S(n)\text{poly}(n)$ bit time. In particular, a quasi-polynomial bound on $S(n)$ gives a quasi-polynomial bound on this composed algorithm.

Proof. The planar output theorem gives polynomially many rays per eligible sector. Proposition 3.7, the determinant bit bounds, and the coverage-verification analysis give polynomial bit cost for their exact generation and replay. The fourth hypothesis supplies the remaining polynomial work per ray. The other hypotheses bound all visited states and justify the two recognition conclusions. Summing these costs over the entire search yields the stated bound. $\square$

The fourth hypothesis is not shorthand for expanding every normal disc. Classical interval-orbit methods are polynomial in compressed input parameters and can handle exponential normal weight [7]. The actual component predicate, its boundary information, and its proof representation must satisfy the same encoded-size accounting. Likewise, efficient verification of an essential hierarchy does not automatically give an efficient method for discovering it.

If a strategy has polynomially many choices per level, a proved $O(\log n)$ depth leads to $n^{O(\log n)}$ states. Depth $O((\log n)^2)$ instead leads to $n^{O((\log n)^2)}$. A bound stated in the size of the latest triangulation is insufficient if intermediate triangulations can grow without a bound in the original n . A progress measure must also pay for discarded suffixes and revisits, not only for successful cuts.

9.1 Research questions with concrete success criteria

The most useful next work should attack an unproved hypothesis or remove an observed bottleneck. The following questions are arranged approximately in that order.

1. Constructive coverage by a small family of sectors. Can the search, under the actual knot-exterior hypotheses, generate a quasi-polynomial family guaranteed to contain an essential-disc standard ray whenever one exists? A useful theorem must give the selection procedure, its encoded cost, and a reason that some witness survives. An existential small support for a convenient example is insufficient: layered meridians already show why support cardinality alone is not a reliable universal parameter. A first manageable target is a natural restricted class of exteriors with a certified reduction to sectors of effective cone dimension three.

2. Certified effective dimension above raw nullity three. Many raw kernel directions never meet the nonnegative orthant. Can the active-face certificates of the recent incoming work identify all coordinates that vanish on the feasible cone, then produce a basis of its exact linear span without enumerating high-dimensional candidate rays? Once that span has dimension at most three, the present geometry applies in it. The missing implementation should certify every removed direction and preserve all feasible points. Compare its cost with simply handling the original sector; do not use a heuristic feasible support as proof of a forced zero coordinate.

3. Incremental exact matching kernels. The sparse constructor eliminates the dense zero-label allocation, but retains dense rational elimination. Nearby sectors differ by a few selected types. Can one maintain a fraction-free factorization, cycle basis, and source-bound rank certificate under those changes? A successful result should bound coefficient growth and update cost, and should remain useful when a changed label splits a previously contracted component.

The benchmark’s source-reuse improvement supplies a baseline; updates must be compared against that stronger baseline.

4. Output-sensitive exact planar geometry. The present half-plane construction is deliberately elementary and has a cubic worst-case arithmetic bound. Can power-diagram construction and an exact segment-overlay method reduce the geometric stage toward $O(k^2 \log k)$, plus the unavoidable cost of writing tR native coordinates? A proof must count clipping by the common source polygon without charging its boundary once per irrelevant equality. It must retain exact behaviour for duplicate forms, lower-dimensional minima, coincident edges, and a degenerate feasible section.

5. Genuine quadratic output or a stronger geometric bound. The abstract grid of Section 5 has $\Theta(p^2)$ rays, but its potential coefficients are not a triangulation-realization theorem. Does there exist a growing family of valid compact knot exteriors or solid tori with matching nullity three and $\Omega(k^2)$ nonlink standard rays in one sector? Alternatively, does the sparse face-incidence structure force a smaller bound? Either outcome would sharpen the parameter that the search should optimize. The two-cap family settles the single-group linear constant, not this multi-group realization question.

6. Structural control of the active groups. The explicit bound separates $A = \sum_v (m_v - 1)$ from the cross-group term. Is it possible to simplify a triangulation, choose a sector, or normalize a cut so that few vertex groups have nontrivial minimum diagrams? One nontrivial group removes the quadratic crossing term entirely. The quantity to preserve is complete witness coverage, not merely the observed number of cells in easy examples. A candidate invariant should include the cost and topological legality of the simplification.

7. A controlled extension to matching dimension four. For $d = 4$, the projective section is three-dimensional and minimum cells are convex polyhedra. Which diagram parameters give a useful output bound for their common refinement? A general all-equalities arrangement is again too coarse. Any extension should distinguish fixed dimension from dimension growing with n , handle non-simple intersections, and provide a checkable coverage representation with a bit-complexity proof. Polynomiality for each fixed dimension would still leave the global sector-selection problem open.

8. Smaller certificates and a formally checked geometric core. The current verifier reconstructs an uncontracted source model for independence and can therefore be slower than the producer. Can sparse elimination be certified by local row identities, rank witnesses, and forest-potential equations, while retaining independence from the producing algorithm? Another useful project is to formalize the canonical-lift correspondence, planar Euler bound, and area-coverage theorem in Lean or another proof assistant. The acceptance criterion must rule out artificial cell subdivisions as well as missing cells; area alone without the distinct-label and dominance conditions does not suffice.

9. Source-preserving compressed topology transitions. After a useful surface is chosen, can cutting and continuation retain all ordered boundary attachments, coorientations, and pattern data with polynomial encoded size? The hierarchy manuscript gives relevant compressed-cutting and certification theorems [9]. The implementation milestone should be one complete transition

with independently replayed source and target objects. Matching an aggregate boundary profile is insufficient when different attachments produce different topology.

10. A global amortized measure compatible with restarts. Can local batched descent, bounded trace search, and selected normal cuts be charged to a single well-founded quantity with only quasi-polynomially many values? The value must account for regluing, unsuccessful local searches, and invalidated later hierarchy stages. A lexicographic genus decrease proves termination only after every coordinate bound and every possible length have been quantified. This is a principal place where a genuine global asymptotic theorem could emerge.

11. Integration under a measured dispatcher. Which exact predicate should select the old specialized rank-two envelope, the new planar method, and the general fallback? The new dimension-three path has a clear coverage advantage; the lower-dimensional controls show why a universal replacement is unwarranted. A useful dispatcher experiment should retain cold-start cost, preparation reuse, coverage replay, and the full recognizer’s time to a justified verdict. It should include both unknots and nontrivial controls and report every capped case.

12. Search-independent benchmark design. Can a benchmark suite be assembled before tuning the selection policy, with independently regenerated complete ray oracles where feasible and structural families beyond them? Hold out source families, not just individual supports of the same triangulation. Report distinct surfaces as well as repeated sector occurrences. Measure bit lengths, number of active cells, overlay intersections, component-query work, memory, and complete-exhaustion time. The present package supplies such measurements for a local stage, but does not estimate the frequency of these sectors in an unbiased distribution of hard knot diagrams.

9.2 A concrete next milestone

The next integration step can be made precise: combine a source-bound exterior, certified effective-face reduction, the appropriate complete envelope enumerator, and one compressed topology transition, with all intermediate certificates replayed by separate checkers. Measure the whole step and preserve the stronger incumbent branches. In parallel, seek a theorem bounding how often that step must be revisited. The first task makes the local theorem operational; the second determines whether it advances the global quasi-polynomial programme beyond a faster exponential search.

10 Reproduction, package contents, and integration

The package is self-contained for its default exact tests and coverage replay. It contains the full article source, rendered PDF, vector figures, the new runtime modules, a pinned snapshot of their maintained dependencies, focused tests, research drivers, a frozen complete oracle, raw experimental evidence, and an additive patch against the pinned repository commit.

Path	Contents and purpose
<code>article/</code>	Main TeX, included sections, bibliography, figures, tables, PDF, and figure-generation script.
<code>code/</code>	Standalone native package snapshot, research drivers, selected fixtures, and the 39-test reproduction set.
<code>fixtures/</code>	Unchanged frozen complete discovery corpus from the preceding minimum-envelope report.
<code>reference/</code>	Unmodified optimized rank-two envelope module used as the stronger performance control.
<code>evidence/</code>	Full corpus results, compact summaries, coverage certificates, replay outputs, family component certificates, raw timings, and regression logs.
<code>scripts/</code>	Independent genuine/abstract differential audit and sparse-constructor benchmark.
<code>integration.patch</code>	Additive repository changes; unchanged dependency snapshots are not part of the patch.
<code>MANIFEST.json</code>	File sizes, SHA-256 digests, and origin classification for every packaged file except itself.
<code>reproduce.py</code>	Manifest verification and isolated reproduction commands.

From the package root, run

```
python reproduce.py
```

This checks the manifest, runs the 31 new focused tests and eight maintained sector-integration tests, and independently replays the thirteen corpus and four family coverage certificates. It writes fresh output into a new reproduction directory. It does not overwrite the supplied evidence.

The longer computations are explicit options:

```
python reproduce.py --audit --family --differential
python reproduce.py --fresh-regina
python reproduce.py --benchmark --sparse-benchmark
python reproduce.py --pdf
```

The first command repeats the full frozen-oracle audit, native family census, and additional differential checks. The second also regenerates the designated complete Regina lists; it needs Regina 7.4 or a compatible installation. The third repeats the paired timing protocol, including resource-limited incumbent runs. The fourth compiles an isolated copy of the article using `latexmk` and `pdflatex`. Optional figure regeneration uses Matplotlib. The standard-library runtime checks do not require either Regina or Matplotlib.

All raw experiment reports are preserved as produced, including their original source paths and timestamps. The executable drivers accept explicit paths or locate their own package root; they do not depend on those historical absolute paths. The complete-output digest is a deterministic correctness target. Wall-clock times, hardware behaviour, and PDF metadata are not expected to reproduce bit-for-bit. The manifest distinguishes the new proposal from unchanged source copied for standalone use.

The patch adds the sparse constructor, planar producer, independent checker, focused tests, and research drivers beneath `Topology/UnknotRecognition/fast/`. Before applying it to a later revision, review any changes to the native source validator, kernel representation, integer codec, and disc-certificate API. The package’s patch was checked against its exact base. It does not replace the maintained dispatcher or assign a new numbered report in the repository’s editorial structure. The TeX article and evidence can be filed according to the repository’s intake process.

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